WORKED SOLUTION

STEP II 2009, Question 11

Mechanics • Power and Resistance

STEP II 2009, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q11

The key idea

Treat the whole train as one body of mass $(n+1)M$ with total resistance $(n+1)R$. At power P the driving force at speed v is P/v, so Newton's second law gives the acceleration straight away. To find a *time* to reach a speed, write the acceleration as $\mathrm{d}v/\mathrm{d}t$, separate the variables and integrate. The last part is an energy balance: the engine's work goes into kinetic energy and into work against resistance.

Solution

Model the engine and trucks as particles moving together (the couplings stay taut and rigid), so the train is a single body. Write

$$m = (n+1)M, \qquad k = (n+1)R$$

for its total mass and total resistance.

The acceleration

Power is force times speed, so at maximum power and speed v the engine's tractive force is P/v. Resolving along the track, Newton's second law for the whole train gives

$$ma = \frac{P}{v} - k, \qquad \text{so} \qquad a = \frac{P-(n+1)Rv}{(n+1)Mv}.$$

The time to reach speed V

Put $a = \dfrac{\mathrm{d}v}{\mathrm{d}t}$ and separate the variables:

$$\frac{\mathrm{d}t}{\mathrm{d}v} = \frac{mv}{P-kv}, \qquad T = m\int_0^V \frac{v}{P-kv}\,\mathrm{d}v.$$

To integrate, split the fraction by writing the numerator in terms of the denominator, $v = \frac{1}{k}(P - (P - kv))$:

$$\frac{v}{P - kv} = -\frac{1}{k} + \frac{P}{k} \cdot \frac{1}{P - kv}.$$

Hence

$$T = m\Big[-\frac{v}{k} - \frac{P}{k^2}\ln(P - kv)\Big]_0^V = -\frac{mV}{k} - \frac{mP}{k^2}\ln\Big(1 - \frac{kV}{P}\Big).$$

Since $m/k = M/R$ and $m/k^2 = M/((n+1)R^2)$,

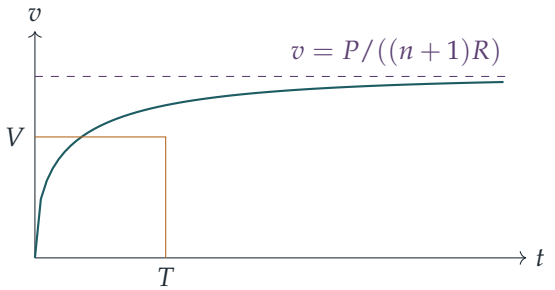
$$T = -\frac{MV}{R} - \frac{MP}{(n+1)R^2}\ln\Big(1 - \frac{(n+1)RV}{P}\Big).$$

Ask yourself Why must P exceed $(n+1)RV$, rather than this just being a condition for the logarithm to make sense?

Physically, the acceleration $(P - kv)/(mv)$ is positive while $v < P/k$ and would be zero at $v = P/k$. The train starts at $v = 0 < P/k$ and speeds up, but the acceleration dies away as v approaches P/k. The formula shows that it never gets there: as $V \to P/k$ from below, $\ln(1 - kV/P) \to -\infty$, so $T \to \infty$. So P/k is a terminal speed that the train approaches but never reaches in finite time, and speeds at or above it are never reached. Therefore speed V can be reached only if $V < P/k$, that is,

$$P > (n+1)RV,$$

as required. (Equivalently, $P \le kV$ would mean the engine at full power can't even balance the resistance at speed V.)



(The sketch is qualitative: the speed rises steeply at first, where the driving force P/v is large, and levels off towards the terminal speed.)

Part (i)

Let $x = \dfrac{(n+1)RV}{P} = \dfrac{kV}{P}$, which is small. Then $\ln(1-x) \approx -x - \frac{1}{2}x^2$, and using $V = Px/k$,

$$T = -\frac{mV}{k} - \frac{mP}{k^2}\ln(1-x) \approx -\frac{mP}{k^2}x + \frac{mP}{k^2}\Big(x + \tfrac{1}{2}x^2\Big) = \frac{mP}{2k^2}x^2 = \frac{mP}{2k^2} \cdot \frac{k^2V^2}{P^2} = \frac{mV^2}{2P}.$$

The first-order terms cancel exactly, which is why we needed the x^2 term. Hence

$$PT \approx \tfrac{1}{2}(n+1)MV^2.$$

Interpretation. PT is the work done by the engine in time T, and $\tfrac{1}{2}(n+1)MV^2$ is the kinetic energy gained by the train. So when $(n+1)RV$ is small compared with P (the resistance force times the speed is small next to the engine's power), almost all of the engine's work goes into kinetic energy and very little is lost to resistance: the train behaves as if there were no resistance.

Part (ii)

In time T the engine, working at constant power P, does work PT. Over the distance X the constant total resistance $(n+1)R$ does work $-(n+1)RX$; in other words the train does work $(n+1)RX$ against it. The train starts from rest and ends with speed V, and the track is horizontal, so there is no change in potential energy. By the work–energy principle,

$$PT = \tfrac{1}{2}(n+1)MV^2 + (n+1)RX.$$

Rearranging,

$$X = \frac{2PT - (n+1)MV^2}{2(n+1)R},$$

as required. This matches part (i): if resistance is negligible, the $(n+1)RX$ term drops out and we are left with $PT \approx \tfrac{1}{2}(n+1)MV^2$.

Checks

- Take $n = 3$, $M = 2$, $R = 1$, $P = 40$, $V = 6$ (so $(n+1)RV = 24 < 40$). The formula gives $T \approx 6.3258$, and then $X \approx 27.258$. Stepping the equation of motion forward in time numerically gives the same values to four significant figures.
- Dimensions: MV/R is (mass)(speed)/(force), which is a time, and so is MP/R^2; the logarithm is dimensionless.
- As $V \to P/\big((n+1)R\big)$, T grows without bound (for the numbers above, $T \approx 28$, 72, 118 at 90%, 99%, 99.9% of the terminal speed), which confirms the condition.
- For small x, the ratio $PT/\tfrac{1}{2}(n+1)MV^2$ is about $1 + \tfrac{2}{3}x$, tending to 1, as part (i) predicts.

Common slips

- Using the mass and resistance of the engine alone instead of the whole train: the engine pulls all $n+1$ vehicles.
- Writing $a = \mathrm{d}v/\mathrm{d}x$, or using $v\,\mathrm{d}v/\mathrm{d}x$, when a *time* is wanted. That route gives distance, not time.

- Expanding the logarithm only to first order in part (i): the x terms cancel, so you get $T \approx 0$ and lose the answer.

Beyond STEP

The equation $m\,\mathrm{d}v/\mathrm{d}t = P/v - k$ is a first-order nonlinear ODE that happens to be separable, and the terminal speed P/k is a stable equilibrium that solutions approach but never reach in finite time. This behaviour is typical near equilibria of smooth ODEs, as uniqueness of solutions shows. Part (ii) is the work–energy theorem, which becomes a central organising principle in university mechanics. For more, see Kleppner and Kolenkow, *An Introduction to Mechanics*, and for the ODE side Boyce and DiPrima, *Elementary Differential Equations*.

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